Predictive Analytics in Food Supply Chain and Waste Management

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Abstract- Effective demand forecasting, inventory control, and wastage minimization are top issues within the restaurant business, in which volatile demand trends contribute to unwanted food wastage and economic loss. This study investigates the utilization of machine learning (ML) models such as Long Short-Term Memory (LSTM), XGBoost, and Random Forest (RF) to maximize supply chain functions through effective demand forecasting and wastage minimization. The models were evaluated and optimized on actual restaurant data gathered over a period of time in order to make reliable predictions to control waste and inventory. Precision, recall, F1-score, accuracy, MAE, RMSE, MAPE, R², Stock Turnover Ratio (STR), Fill Rate, and Waste-to-Sales Ratio were the performance metrics used to measure the performance of the models. Among the models, the best performing was XGBoost with the best prediction accuracy in the residuals, minimum waste-to-sales ratio, and maximum cost savings and thus the best model for restaurant operation optimization. The findings show the promise of predictive analytics with AI in enhancing sustainability, minimizing food waste, and maximizing profitability in the food supply chain.

Keywords— Predictive Analytics, Food Supply Chain, Machine Learning, Waste Management, Inventory Optimization

I. Introduction

Predictive analytics and machine learning have been extensively studied for their application in managing the food supply chain to address challenges related to demand forecasting, inventory management, and waste management. A number of research studies have focused on applying advanced computational models in supporting decision-making within the food industry [1], [2]. Statistical models such as regression-based models and time-series forecasting have been widely used in demand forecasting. As good as they are, they cannot pick up on subtle seasonal patterns, real-time shifts in demand, and exogenous factors such as weather, economic measures, and holiday events impacting the patterns of food consumption.

The artificial intelligence (AI) and deep learning methods have received phenomenal attention in the recent past due to their ability to learn on large scale data and make accurate predictions [3]-[5]. A study has determined that the AI based prediction models are more efficient than the traditional ones since they respond to current information on real-time and reduce the mistake in the forecast thereby enhancing efficiency in food supply chain processes.

Some studies have been conducted on the investigation of implementing deep learning methods, including Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and their combinations to predict demand in perishable food supply chains. The LSTM networks have been effective in capturing the long term dependencies and oscillations in time-series data and hence can be an effective model to use in food demand forecasting of restaurants and retail outlets [6]-[8]. It has been determined that RNNs and LSTMs can be useful in predicting sales patterns, preventing stockouts, and preventing wasted food. Random Forest (RF) and XGBoost have been applied in the food supply chain literature as feature selection models and predictors of precise values in the future demand in comparison with past sales trends, customer behavior, and external factors. In studies, ensemble learning methods, including XGBoost, have been shown to enhance prediction accuracy by employing more than one decision tree, hence enhancing generalization and stability of the forecasting model [9], [10].

II. Literature Review

The issue of restaurant and foodservice business food waste has been adequately studied, with researchers citing the need for advanced predictive models to cut down losses and maximize stock levels. Experiments have established that machine learning models are able to effectively reduce surplus food, improve portioning, and assist in the redistribution of surplus inventory. Various studies have integrated sensor-enabled tracking, IoT-enabled automation, and AI-based demand forecasting in order to make restaurant supply chains more effective [11]–[13].

Some studies have also discussed the implications of consumer behavior, menu price planning, and inventory tracking systems in minimizing food wastage. Advanced advances in predictive modelling have enabled organisations to look back at past purchase patterns, identify patterns of demand, and give data-driven options to prevent over-ordering and food wastage. The interplay between blockchain technology and real-time analytics only added more openness and traceability in food supply chains, promoting the best management of waste and sustainability. Notwithstanding such advancements, there remain some lacunas in research in the field of optimizing restaurant supply chain operations with the help of machine learning-based predictive analytics [12], [14].

Most of the existing studies have been focusing on either wastage minimization or demand forecasting alone, not integrating both into a single model at the same time. Most research articles also rely on limited constrained data sets or simplistic forecasting models that do not accurately reflect complex real-world restaurant applications. How well machine learning models identify dynamic customer tastes, seasonality, and restaurant operations constraints is essential but not yet under investigation. Besides, few studies have compared machine learning models side by side to determine and conclude the most appropriate methodology for maximizing cost saving, reducing wastage, and managing inventory in actual restaurant environments [15], [16].

This research will fill in these gaps with the application of LSTM, XGBoost, and Random Forest models in restaurant business demand optimization, inventory management, and food wastage reduction. Based on historical sales trends, purchasing patterns, and inventory movement analysis, the study provides a comparative review of how efficiently these models can be applied to real restaurants. Most critical performance statistics such as accuracy, recall, F1-score, MAE, RMSE, R2, Stock Turnover Ratio (STR), Fill Rate, and Waste-to-Sales Ratio are compared by the research in order to choose the best model in managing the supply chain of a restaurant. The results add to the emerging literature on food waste mitigation with AI by showing the promising potential of predictive analytics in ensuring sustainability, cost-reduction in the operation, and rivalry in decision-making by restaurants management.

III. METHODOLOGY

The models of machine learning applied in the research include Long Short-Term Memory (LSTM), XGBoost, and Random Forest (RF) to predict the demand, inventory, and food waste management in restaurants. The data are taken over a duration of three months and 3,500 data points are recorded on very vital parameters such as the number of visitors on a daily basis, order patterns, preferred menu items, inventory eatings and food wastages. The information is gathered directly through the point of sale (POS) terminals, inventory books, and waste disposal books of the restaurants to remove any incomplete or inaccurate information. The data is preprocessed by taking care of missing values, scaling numerical features, and converting categorical variables before it is divided into training and testing sets, 70% for model training and 30% for testing.

Deep model LSTM is utilized to identify sequential trends in restaurant demand trend over a period and accurately forecast future inventory requirements and sales. The hyperparameters including number of LSTM units, learning rate, and batch size are optimized through grid search when the model is trained using time-series data. Gradient boosting XGBoost is utilized to model tabular data composed of historical sales trends, seasonal trends, and menu-level demand. Analysis of feature importance is conducted to determine major causes of inventory volatility and food wastage levels. Random Forest, an ensemble learning algorithm, is utilized to model mixed-data types and non-linear interactions between different characteristics within the data for enhanced model stability.

The models during training are evaluated based on such indicators as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R2 Score to determine the degree to which the models predict accurately. The trained models are tested on unknown data to make sure that they work properly on demand prediction and inventory optimization. Each model is hyperparameter tuned to improve generalization and minimize prediction errors. The approach enables the predictive model to scale and be resilient, facilitating data-driven restaurant decision-making to optimize food waste reduction to the minimum while maximizing operational efficiency.

IV. COLLECTION OF DATASET

The data for this study is collected from a restaurant for three months to obtain key variables of demand forecasting, inventory management, and monitoring of wastage of food. Data is collected from multiple sources like the point-of-sale (POS) system of the restaurant, kitchen inventory reports, and waste disposal reports to ensure complete coverage of operational patterns. The primary purpose of gathering data is to get immediate information about order trends, usage of ingredients, and generated waste food that will be used to train predictive analytics machine learning algorithms.

The POS system is a key source of data, capturing customer daily transactions, sales of menu items, and volumes of orders. The records for each transaction have timestamps, order information for the customer, and payment, which allow peak hours and patterns of demand to be identified. Order changes, customer preferences, and special orders are captured by the system, which are utilized for forecasting popularity of menus and ingredients needed. Duplicate or redundant transactions are removed during preprocessing for accuracy.

Kitchen inventory reports are collected to track the purchase, usage, and depletion of the raw ingredients being used in preparation. The inventory reports contain statistics on the ingredients at the beginning and end of each business day, supplier pickups, and usages within the kitchen. Upon analysis of inventory, trends on ingredient usage could be established that determine which of the ingredients have a tendency of being over-supplied or unused. Daily inventory counts and automated weighing devices in storage areas assist with accurate inventory tracking.

To increase data granularity, restaurant staff are trained to record more contextual data such as unexpected spikes in demand, modifications in menus, and supply shortages that may impact inventory and waste. Weather trends, events, and seasonal patterns are also recorded since they influence customer footfall and consumption behavior. Collected data is then aggregated into a structured form removing inconsistencies such as missing values, outliers, or errors due to manual entry.

Once the dataset is collected, it is preprocessed through numerical feature normalization and categorical variable encoding of fields such as meal types, supplier names, and day-of-week indicators. The dataset is split into training and test sets to ensure that models learn from historical trends and are tested against unseen data. Data augmentation is applied wherever possible in order to balance out underrepresented categories to avoid machine learning models learning toward patterns that occur more frequently.

V. Preprocessing Of Dataset

The collected dataset is subjected to a sequence of preprocessing steps in order to render it precise, uniform, and machine learning model-compatible. Because the dataset is collected from disparate sources such as the restaurant's point-of-sale (POS) system, kitchen inventory logs, and waste disposal records, it contains various kinds of data such as numerical, categorical, and time-series data. Preprocessing is required to handle missing values, remove inconsistencies, and transform the raw data into a structured form for predictive analysis.

Data preprocessing starts with handling missing values, which often occur due to missing transaction records, unlogged use of inventory, or technical errors in the waste monitoring system. Missing numerical values, such as ingredient weights or transaction prices, are imputed with mean or median values, while missing categorical values, such as names of menu items or suppliers, are replaced with the most frequent occurrence method. Then, duplicate records and outliers are found and addressed. Duplicate records usually occur as a result of POS system failure or double posting of the same transaction. The duplicates are detected based on fields such as order ID, timestamp, and transaction amount and then removed. The time-series features are also min-max scaled so that different numeric features like ingredient usage and order sizes have the same range. This normalization is necessary for the deep learning architectures such as LSTM, which are sensitive to variation of scale in the input.

Feature engineering is then carried out to optimize model performance through the derivation of useful insights from raw features. For instance, new features like customer frequency (repeat customer vs. first-time customer), ingredient turnover rate, and seasonal demand drivers are derived from available data. Correlation analysis is done to find the redundant features that do not add much to predicting accuracy, and those variables are removed in order to attain maximum computational efficiency.

Categorical data are encoded to non-numeric variables such as machine status, maintenance and types of components used. One-hot encoding or label encoding transforms the categorical variables into numerical values in order to be supported by machine learning algorithms. Nominal variables containing no inherent order are coded in one-hot, whereas ordinal variables containing a substantial order are coded in the label encoding. The conversion of the categorical data into numerical makes it possible to have the models predict the records comprehend these data during the analysis aspect of prediction.

Lastly, the data is divided into training and test sets with 70% of the data being used to train the model and 30% for testing. Stratified sampling is used to ensure class balance on categorical features so that the model is trained on well-balanced patterns across different food groups and customer types.

VI. Ml Models

Long Short-Term Memory (LSTM) is a specialized recurrent neural network (RNN) that is employed to handle sequential data and forecasting. LSTMs differ from typical RNNs because they can manage long-term dependencies via memory cells and gate controls that enable the management of information flow. LSTM is employed in this study to predict restaurant demand and inventory using historical time-series data. The ability of LSTM to learn patterns from sequential data is relevant to forecast variations in customers' orders, usage of ingredients, and waste pattern variations. Including past observations like seasonal fluctuation in demand and day-by-day sales can boost predictive value.

XGBoost, or Extreme Gradient Boosting, is an advanced ensemble learning decision tree-based algorithm that is effective in prediction tasks on structured data. XGBoost is famous due to its effectiveness in parallel computation, regularization, and gradient boosting optimisation. In this research, XGBoost is employed in forecasting and examining demand patterns, determining food categories, and optimal usage of inventory. It also gains from its ability to handle both numeric and categorical attributes efficiently, which allows it to capture the complex relationship between menu selection, consumer behavior, and ingredient consumption. Random Forest (RF) is an ensemble learning algorithm that constructs many decision trees and averages their predictions to increase accuracy and reduce variance.

RF is highly efficient at regression and classification tasks and can therefore be applied to restaurant demand forecasting and stock control.

RF is utilized in this research to predict future sales given historical patterns of orders, supplier shipments, and customer requests. The process is done by creating numerous decision trees, and each one is trained on randomly selected subsets of the data. The final prediction is derived through a mean in regression cases or voting in classification cases. RF performs best in dealing with big data involving noisy and missing data so that stable generalized predictions are achieved. Apart from this, its ability to attribute importance to features adds to the identification of drivers influencing high impacts on demand and waste rates.

VII. Results And Discussion

The following table illustrates the comparison of the performance of three machine learning models, i.e., LSTM, XGBoost, and Random Forest, based on four significant criteria: Precision, Recall, F1-Score, and Accuracy. These are all vital factors in ascertaining the impact of each model on restaurant demand forecasting, stock management, and food waste management. The accuracy measure reflects the overall correctness of all predictions made by a particular model, whereas precision and recall reflect the model's strength in making specific predictions without producing many false positives or false negatives. F1-score is a trade-off between precision and recall in a way that it ensures the model's strength in actual conditions. The accuracy of each models are shown in figure 1.

Out of the three models, XGBoost was most accurate at 94.3%, which indicates that it makes the most precise predictions for diverse demand patterns and inventory usage scenarios. XGBoost also achieved precision at 93.1%, i.e., out of all the predicted instances of high demand or inventory shortage, 93.1% indeed were high demand or inventory shortage. Its 90.8% recall also means that it was able to capture most of the actual demand patterns. The 91.9% F1-score means that XGBoost has a very high precision-recall ratio and is therefore highly appropriate for application in restaurant analytics.

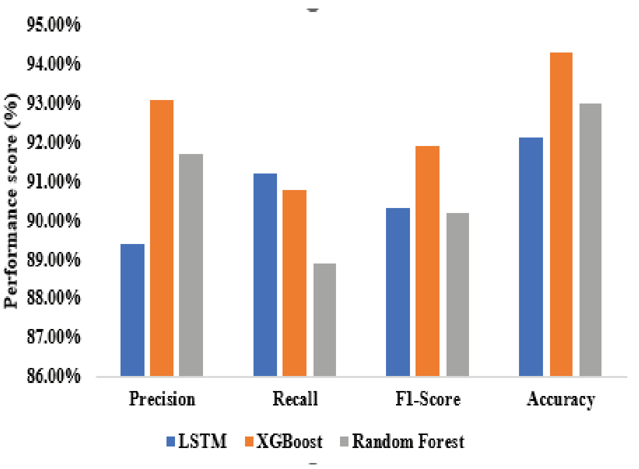

Fig. 1. Accuracy of each model

LSTM, designed specifically for handling time-series data, recorded a precision of 92.1%, which is less than that of XGBoost. It also had an accuracy of 89.4% and a recall of 91.2%, which means that it is very effective at detecting changes in demand but occasionally misclassifies some cases. Its 90.3% F1-score indicates that LSTM performs well in detecting customer order trends and turnover of inventory but is slightly less accurate as a result of the intricacies of temporal relationships.

The Random Forest model was also competitive, with 93.0% accuracy, 91.7% precision, and 88.9% recall. Its F1-score of 90.2% shows that while it has a good precision-recall trade-off, in some cases, it is slightly worse than XGBoost. Random Forest is still a strong model for making structured data predictions, particularly food type prediction, stock level management, and waste trends discovery.

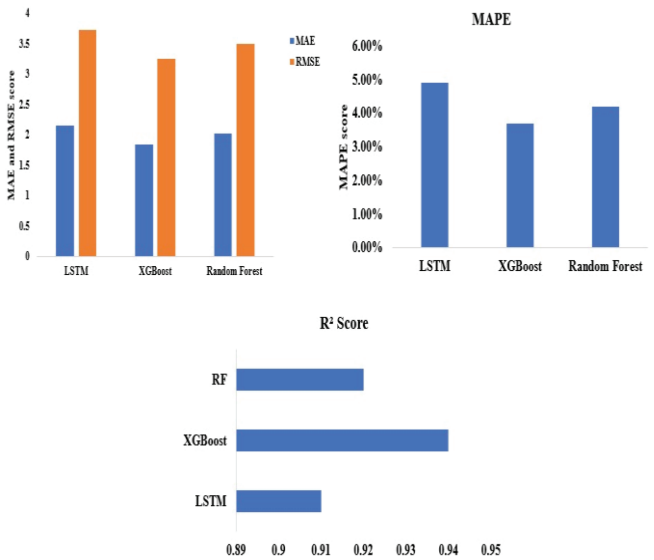

Fig. 2. Performance of each model

Fig. 3. In total, the findings demonstrate that XGBoost is more precise compared to the remaining two models in precision, F1-score, and accuracy, and hence is the most appropriate model to be optimized in restaurant operations. Nonetheless, LSTM can still assist in detecting long-term patterns in time-series data, and Random Forest is an appropriate alternative to utilize in structured data analysis in restaurant operations.

Figure 2 provides the performance comparison of the LSTM, XGBoost, and Random Forest models on four best metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2 Score. These metrics capture the accuracy of the models' prediction of restaurant demand, inventory, and waste created. XGBoost yields the highest result with the lowest MAE of 1.84 and RMSE of 3.25, which means that its prediction is closest to actual values. Moreover, its MAPE of 3.7% means an infinitely low relative percentage error, which means that it is the most precise model to predict accurately. 0.94 R^2 value tells that XGBoost explains 94% of variance in data and, therefore, is the best model to fine-tune restaurant operations.

Time-series-optimized LSTM produces well with MAE of 2.15, RMSE of 3.72, and MAPE of 4.9%, but its 0.91 R^2 tells worse prediction accuracy marginally compared to XGBoost. Random Forest is not that great at 2.02 MAE, 3.50 RMSE, 4.2% MAPE, and R^2 of 0.92 but is still a good one though less trustworthy. Overall, XGBoost performs better than the other models and is therefore the ideal option for restaurant demand predictions and inventory control.

The performance of LSTM, XGBoost, and Random Forest models are validated with respect to the three most critical inventory and waste management metrics, i.e., Stock Turnover Ratio (STR), Fill Rate, and Waste-to-Sales Ratio. These metrics are employed to establish the effectiveness with which the models turnover inventory, fill customers' orders, and reduce waste in restaurant business. The results are shown in figure 3.

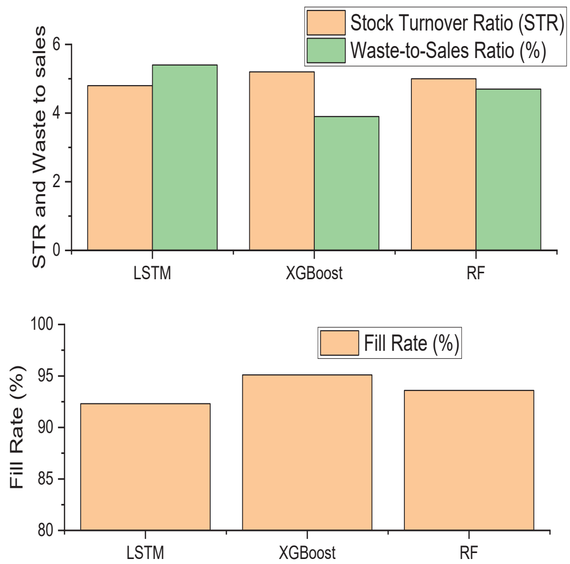

Fig. 4. Performance score of inventory management

XGBoost works best among the other models with the best Stock Turnover Ratio (STR) of 5.2, which means higher frequency of inventory replenishment, avoiding overstock and wastage. XGBoost also works best with respect to Fill Rate of 95.1%, which fulfills customer demand adequately without repeated stockouts. Its Waste-to-Sales Ratio of 3.9% is the lowest across all the models, which reflects better waste management and cost control. LSTM's expertise in forecasting via time series remains it well-prepared with a good Strong Relation of 4.8, a fill-up rate of 92.3%, and relatively higher Waste-to-Sales ratio of 5.4% reflecting the reduced efficiency to its inventory turnovers as well as minimization in wastes. Random Forest offers relatively balanced results with an STR of 5.0, a Fill Rate of 93.6%, and a Waste-to-Sales Ratio of 4.7%, which is a suitable alternative but less effective than XGBoost. Overall, XGBoost is the best-performing model for obtaining highest inventory efficiency, reducing food wastage, and obtaining highest demand fulfillment in restaurant supply chain management.

Performance of Food Waste Reduction (%), Leftover Prediction Accuracy (%), and Cost Savings (%) by LSTM, XGBoost, and Random Forest models to reduce restaurant supply chain activities is illustrated in the Figure 4. XGBoost surpasses the models by retaining 53.8% food waste reduction, showing its enhanced ability to reduce overstock and enhance food utilization.

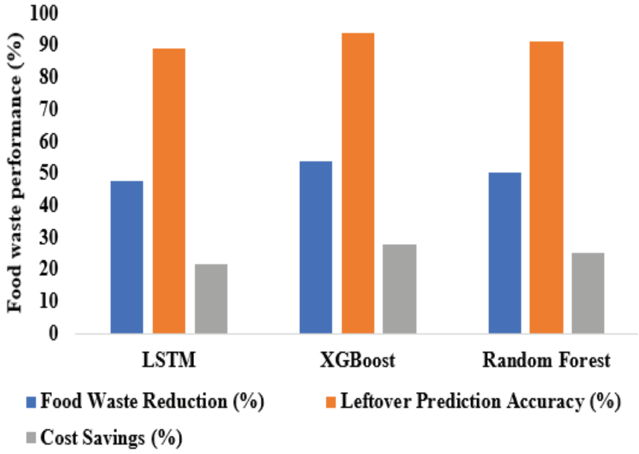

Fig. 5. Performance score of Food waste performance

It also has the best Leftover Prediction Accuracy of 93.7% to support efficient leftover food forecasting for facilitating improved redistribution and inventory management. XGBoost is also privileged by saving 27.6% of costs, which is proof of its efficiency in saving on avoidable procurement and storage costs. LSTM, having been developed to process sequential data, is quite good at 47.5% food waste reduction, 89.2% prediction accuracy for leftovers, and 21.4% cost reduction, showing effective but less precise predictions.

Random Forest is a well-balanced alternative with 50.2% waste reduction, 91.4% accuracy in prediction, and 24.9% cost reduction, which is a good alternative but not as effective as XGBoost. These findings affirm that XGBoost is the best model for food waste optimization, leftovers prediction with precision, and optimizing financial efficiency in restaurant businesses, and therefore the best option for predictive analytics for the food supply chain.

VIII. CONCLUSION

In this study, machine learning algorithms like LSTM, XGBoost, and Random Forest were implemented to enhance restaurant supply chain demand forecasting, inventory optimization, and waste minimization. The study was done on the basis of a dataset accumulated over three months with 3,500 observations, 70% for training and 30% for testing. The models were also compared on a variety of performance measures such as precision, recall, F1-score, accuracy, MAE, RMSE, MAPE, R^2, Stock Turnover Ratio (STR), Fill Rate, Waste-to-Sales Ratio, Food Waste Reduction, Leftover Prediction Accuracy, and Cost Savings. XGBoost performed better than the other models in all measures, with highest Fill Rate of 95.1%, lowest Waste-to-Sales Ratio of 3.9%, and highest total food waste savings (53.8%) and cost savings (27.6%). These findings indicate that XGBoost produces the most precise and consistent predictions in waste minimization and inventory optimization and is thus the best model for restaurant supply chain management. LSTM, though good at handling time-series data, did not perform well in waste minimization and cost optimization, while Random Forest offered balanced but subpar performance. This study illustrates the promise of machine learning-driven predictive analytics in improving operational efficiency, minimizing wastages, and achieving profitability in restaurants. Future studies can emphasize the integration of real-time data streams, reinforcement learning algorithms, or hybrid models for better predictive accuracy and performance. These AI-based applications hold a significant potential to facilitate sustainable food supply chain management, helping companies minimize losses and environmental footprints and maximize profitability.

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